

TOPIC 2.1

How are chemical processes part of our lives?

Key Concepts

- Applications of chemistry are everywhere in the world around you.
- Knowing how to handle chemicals helps keep us and our environment safe.

Curricular Competencies

- Make observations aimed at identifying questions about the natural world.
- Ensure that safety and ethical guidelines are followed in investigations.
- Experience and interpret the local environment.
- Consider social, ethical, and environmental implications of findings.

The link between fireworks and celebration is more than 1000 years old. Its origins are in China, where the earliest fireworks had been invented centuries before. However, colourful displays like this one at Vancouver's annual Celebration of Light festival only became common less than 200 years ago. Before then, the only colours available were those you might see in a typical fire—orange, white, and yellow. Compounds of barium (which supply green hues) and strontium (which supply red) were not available until the elements themselves had been discovered. Strontium was discovered in 1790 and barium in 1808.

Many of today's fireworks use the ionic compound potassium perchlorate, KClO_4 . When ignited by a lit fuse, this compound breaks apart and releases oxygen gas. Carbon and sulfur in the fireworks act as fuels. They react instantly with the oxygen, producing hot gases that provide some of the explosive force to propel the firework upward. The heat of the explosion causes metals and metal compounds in the firework to give off the coloured lights that we look forward to.

Starting Points

Choose one, some, or all of the following to start your exploration of this Topic.

1. **Identifying Preconceptions** In the popular media, as well as in sources of science information, it is common to hear or read the term *chemical*. What do you think of when you hear this word? What does it mean? What other terms could be used to improve clarity and precision when we communicate about chemistry in the world around us?
2. **Applying** To ensure that the fuel burns completely, fireworks are often packed with an excess of KClO_4 . This means that even in fireworks that work as intended, there is often some of the compound left over. However, perchlorate ions can cause health problems such as interfering with thyroid function. As well, the compound dissolves easily in water, which lets perchlorate ions, ClO_4^- , pollute wells and other water sources. Re-read the quotation on page 96. What is green chemistry? What questions do you have about it, and how might it help with the perchlorate problem? How could you begin to search for answers?
3. **Considering First Peoples' Perspectives** Foundations of First Peoples science rest on the several related themes that include interconnectedness, reciprocity, transformation, renewal, and connection with place. Reflect on what you already know about chemistry. Refer also to the Unit at a Glance feature for this unit. At this moment, where do you see the ideas of Western science aligning with those of First Peoples science? Are there ways in which the ideas might have less in common? If so, are there ways to encourage greater harmony?



Key Terms

There is one key term that is highlighted in bold type in this Topic:

- chemical reaction

Flip through the pages of this Topic to find this term. Add it to your class Word Wall along with its meaning. Add other terms that you think are important and want to remember.

CONCEPT 1

Applications of chemistry are everywhere in the world around you.

Activity

Chem Chat

Take a few minutes to list and discuss all the ways you have relied on chemistry today, this week, and in the last month. As a class, compare your lists. What ideas appeared often? Which were less common? Which ideas surprised you?



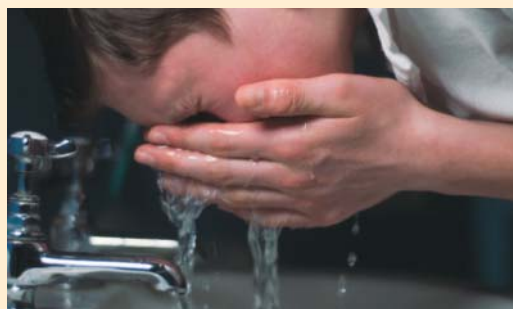
chemical reaction process in which atoms of one or more substances are rearranged to form different substances

Figure 2.1 Chemistry in modern life

Loud explosions, flashes of light, puffs of smoke—these are some of the things you might expect from chemical reactions. Many **chemical reactions** are explosive, colourful, and smoky, but many more take place in the world around you without your even noticing them. Whether they are obvious or not, chemical reactions maintain and sustain your health and the health of all life on Earth. This doesn't mean, however, that chemical reactions and other chemical processes come without hazards. **Figure 2.1** shows just a few of the ways that chemistry affects our lives.



Foods are sources of nutrients that require chemical reactions to break them down into forms that body cells can use. **Evaluating:** How do the chemicals in fresh food, such as an apple, differ from those in prepackaged foods, such as a fruit-flavoured snack? How important are those differences?



Personal care products have been specifically developed with certain properties. Many companies keep their procedures for making them top secret, especially if the product is highly profitable. **Questioning:** What role does an understanding of chemistry play in the production and the use of personal care products?



Chemistry allows doctors to use equipment such as MRI machines to visualize internal structures and processes in the body. **Applying:** Describe a medical advance due to chemistry that has helped you or someone you know.



Frog's Leaf (common plantain) is respected by First Peoples for the properties of its healing chemicals. **Communicating:** What special relationships do First Peoples have with plants in their areas?





Plastics are human-made (synthetic) products that have a wide range of uses. They were invented to be highly stable, so they do not break down easily as they are used.

Analyzing: How is the stability of plastic products both an advantage and a disadvantage?

Natural sources for fabrics we wear are wool, cotton, silk, and linen. The chemical industry has expanded the range and properties of fabrics by inventing synthetic fibres such as nylon, acrylic, spandex, and polyester. **Applying:** What are some advantages and disadvantages of synthetic clothing?



The vehicles that move us from place to place on land, in the air, and on water also contribute to pollution of all three. Our homes and vehicles are safer thanks to chemistry. However, the increased number of them in society makes a significant contribution to air pollution. **Communicating:** What role has chemistry played, both positive and negative, in the development of the transportation industry?

Activity

Chemistry Career Fair

What comes to mind if you think of jobs or careers that involve chemistry? You might think of work that has the word *chemical* or *chemistry* in it, such as chemical technician or chemistry teacher. However, most jobs or careers that involve chemistry do not have the term in their name. Auto body technician, metallurgist, and dental hygienist are three examples.

What others are there? Brainstorm as many examples as you can think of. In each case, record any part or parts of the job that involves chemistry. Who can you talk to or where can you go to find out more about a job or career that might interest you?

Connect to Investigation 2-A on page 114



Before you leave this page . . .

1. Think of what you did before coming to school today. Name three things you did or used that involved chemistry.
2. Is it possible to live a life that is free of chemicals or chemistry? Explain your thinking.

Make a Difference

Chemicals in Our Water

In nature, pure water doesn't exist. Water is called the universal solvent because many different substances can be dissolved in it. These substances can be naturally occurring, such as metals and ions found in the soil, or they can be the result of human activities. In some cases, these human activities are purposeful, such as adding fluoride to municipal drinking water to reduce tooth decay. In other cases, they are the result of household, mining, and industrial wastes, which cause water pollution. These can include human waste, solids such as plastics, potentially toxic chemicals, and pharmaceutical waste that includes prescription medications and personal care products that people flush down toilets or pour down drains.

Water Treatment Facilities Only Do So Much

In most municipalities, waste water that comes from homes, schools, and other buildings enters into a sewer system, which is a system of pipes that delivers the water to a treatment facility. Treatment of waste water varies from one town or city to another, depending on different factors, such as the size of the population and the economic cost of building, operating, and maintaining the facility. Even the most highly efficient municipal facilities cannot remove all polluting chemicals. That means they remain dissolved or suspended in the water that is delivered to people or flushed into local aquatic ecosystems.



British Columbia is home to numerous bodies of water, as well as a diverse and lengthy coastline.

Storm drains are openings covered by heavy metal grates along city streets that collect run-off and excess rainwater. However, storm water goes directly into local lakes and rivers with no treatment. This means any water that runs along the streets, driveways, and parking lots can pick up all types of pollutants and deliver it directly to our waterways. It also means that anything someone dumps into the storm drain will end up in a local lake or river.

Many First Nations in Canada do not have safe drinking water. Some have had to boil their water for many years. While most of these communities have some sort of treatment facilities, they are inadequate, so water supplies continue to be contaminated.



Developing a Plan

1. As a group, decide on how you wish to contribute to improving water quality. Consider the following questions to help you develop your plan:
 - Will it apply to your local community, or is it for a wider region such as the province or another country?
 - What aspect of water quality improvement do you want to address—improving public awareness? combating further pollution? cleaning up already polluted water resources?
 - What type of effort do you want to make? Do you want to provide information and awareness of a particular problem? Do you want to organize a volunteer group to work on a particular issue? Do you want to suggest how each individual can make a difference or how institutions or governments should make a difference?
 - Will you include information about different technologies that are available to help reduce water pollution?

- How will you reach your target audience? How will you convince people that this is an important issue?
 - What information do you need to find out in order to decide on and develop your plan?
 - What criteria will you use to determine if your plan is a success or needs changes to adjust to unforeseen problems?
2. Develop your plan and have your teacher approve it. With your teacher's permission implement your plan or a part of it. This can be done using your suggested target audience or members of the class as a trial run.
 3. Assess how well your plan worked.

Analyze and Evaluate

1. What were some of the challenges you experienced when developing your plan? What were some of the challenges in implementing the plan?
2. Based on what you have learned, how can you, personally, change some of your habits to help address water quality issues? How can community members change their habits? How likely is it that you and/or community members actually will make these changes?

Apply and Innovate

3. Some organizations have websites that include stories of successful projects and people who have made a difference. Compose a summary of how your plan made a difference that could be posted on such a website. If you could not implement your plan, develop a story that conveys the type of success you wished for.

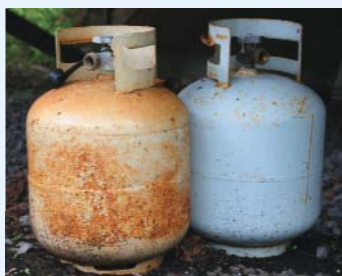
CONCEPT 2

Knowing how to handle chemicals helps keep us and our environment safe.

Activity

Chemical Safety in Our Daily Lives

In a group, discuss possible ways people could be better informed about how to safely handle and dispose of potentially harmful materials such as those shown in the photos.



Chemical Safety in the Science Classroom

Many of the chemical compounds in products that people use at work or that you will use while performing investigations at school have properties that make them potentially dangerous. WHMIS (Workplace Hazardous Materials Information System) labels provide information about chemicals in science labs and at workplaces (Figure 2.2). Safety Data Sheets (SDS) provide further information, which by law must be available for every chemical. This information includes its chemical and physical properties and how it should be handled, stored, and disposed of. SDS also inform about hazards and first aid treatment in case of an accident.

Figure 2.2 WHMIS symbols provide information about each chemical in the lab and workplace.

	Exploding bomb (for explosion or reactivity hazards)		Flame (for fire hazards)		Flame over circle (for oxidizing hazards)
	Gas cylinder (for gases under pressure)		Corrosion (for corrosive damage to metals, as well as skin, eyes)		Skull and Crossbones (can cause death or toxicity with short exposure to small amounts)
	Health hazard (may cause or is suspected of causing serious health effects)		Exclamation mark (may cause less serious health effects or damage the ozone layer)		Biohazardous infectious materials (for organisms or toxins that can cause disease in people or animals)

Safety Rules

You and your classmates are responsible for keeping yourself and others safe while handling chemicals. To do this, you need to know and respect the safety rules. Some important ones are represented in **Figure 2.3**.

You must also read *Safety in Your Science Classroom* on pages xiv–xvii. Your teacher may provide you with further safety rules to follow.

Connect to Investigation 2-B on page 116



Figure 2.3 Working in science classrooms requires you to know how to perform certain tasks safely.

Activity

Minimizing Risks

Working in small groups, choose a profession from the examples shown, or from ideas within your group. In discussion and through research, complete the table below.

Consult WHMIS, SDS, and HHPS information as needed. Combine your information with other groups to create a single summary table.

Profession	What chemicals and/or chemical reactions are used and why?	Risks associated	Ways to minimize risks to personal safety	Ways to minimize risks to the environment

The chemicals used in car restoration and painting can be hazardous to auto body repair technicians.



Welders rely on chemical reactions to perform their jobs—the same reactions that cause explosions.



Firefighters and other first responders not only fight fires but also may be exposed to hazardous chemicals.



Many professionals who do home renovations must work with potentially dangerous products, such as paint solvents, and be aware of possible exposure to harmful materials such as asbestos.



Before you leave this page . . .

1. Describe one action you can take to better handle a chemical so that any personal and environmental hazard can be minimized.

Check Your Understanding of Topic 2.1

QP Questioning and Predicting PC Planning and Conducting PA Processing and Analyzing E Evaluating
AI Applying and Innovating C Communicating

Understanding Key Ideas

1. What do WHMIS and SDS stand for? What are their purposes and how are they associated with each other? **PA E**
2. Describe how to properly heat a test tube using a Bunsen burner. **PC AI**
3. Describe one precaution you should take when you see each of the following. **E C**
 - a) 
 - b) 
4. Name three things to do when preparing to carry out an investigation or activity that involves the use of chemicals. **QP PC**

Connecting Ideas

5. Make a sketch of your science classroom. Include the safety equipment and where it is located. **C**
6. Create a graphic organizer to summarize the positive and negative effects of chemicals introduced in the Topic. **E C**
7. Many farmers apply chemical fertilizers and other substances, such as insecticides, to help maintain their crops and improve the yields. Describe two things the farmer could do to reduce the risk of personal exposure and exposure to others of these chemicals.

AI E



8. You have been asked to develop a safety plan for performing an investigation. **PA AI**
 - a) How do WHMIS symbols help to do this?
 - b) Describe how an SDS could be used.
 - c) Why should chemical disposal be part of your plan?

Making New Connections

9. Iron-containing hemoglobin molecules in red blood cells carry oxygen to the cells of your body. Carbon monoxide binds to iron in hemoglobin molecules 200 times more strongly than oxygen does. When carbon monoxide molecules bind to hemoglobin, they tend to stay bound. Oxygen cannot bind to hemoglobin that is already bound to a carbon monoxide molecule. **QP PA AI C**



carbon
monoxide



oxygen
molecule

- a) Use the models of oxygen and carbon monoxide to infer why a hemoglobin molecule might accept a carbon monoxide molecule as well as an oxygen molecule.
 - b) Carbon monoxide poisoning can lead to effects such as headaches, dizziness, nausea, loss of consciousness, and death. These effects increase with increased concentrations of the compound. Use your understanding of gas exchange to explain why carbon monoxide is toxic. Recall your previous studies about cellular respiration and gas exchange in the body, or aid your recall by consulting suitable information resources.
10. Use your experiences in answering question 9 to pose at least one other question that you could investigate about the cause or effect of toxic chemicals on the body. **QP AI**

Fighting Fires with Chemistry

What's the Issue?

Fire extinguishers are mandatory in laboratories and many workplaces. Homeowners are also strongly encouraged to keep and maintain a fire extinguisher in locations where fires commonly occur, such as in or around kitchens, wood stoves, and garages.

Different Fires Need Different Extinguishers

There are different types of fire extinguishers, and the chemicals they contain have properties that are specific to certain types of fires. Using the right extinguisher will help to put out a fire. Using the wrong one could spread the fire and put lives in greater danger. As shown in the table on the next page, manufacturers put a letter and symbol on each fire extinguisher to show the type of fire it can be safely and effectively used to put out.



Dig Deeper

Collaborate with your classmates to explore one or more of these questions—or generate your own questions to explore.

1. What type of fire extinguisher is present in your science lab? Why is this the correct one to use? Describe how to properly use it.
2. What type of fire extinguisher should you use if your school computer lab catches fire due to an overloaded electrical outlet? Explain your reasoning. (**Hint:** What does the lab contain other than computers?)
3. Why are kitchens and garages common places where fires occur in homes? How can the fires that occur in these places be different? It is recommended that every home have at least one fire extinguisher. Where should they be located? Explain your answers.
4. Halons are types of chemical compounds that have been used traditionally in fire extinguishers.
 - a) Use a fact sheet provided by your teacher or other information resources to find out more about using halon to extinguish fires, as well as the health and environmental risks.
 - b) Canada is currently phasing out the use of halon in fire extinguishers. What is the reason for this? How is halon to be disposed of safely? What will be used in its place?

Class of Fire (Letter and Symbol)	Fuel Source for Fire	Type of Extinguisher	Risks and Benefits
	ordinary combustibles (wood, paper, cloth, and plastic)	- air-pressurized water - dry chemical - chemical foam	- Dry chemical extinguishers should be used with water, because they may not fully extinguish the fire on their own. - Carbon dioxide fire extinguishers should not be used on fires involving ordinary combustible materials; the fire may re-ignite.
	flammable liquids (non-cooking) (gasoline, petroleum oil, paint), flammable gases	- carbon dioxide - dry chemical - chemical foam - halon (a liquefied, compressed gas)	For liquid fires, never use air-pressurized water extinguishers. They can spread the fire and cause burns from splattered fuel.
	electrical equipment (live electricity)	- carbon dioxide - dry chemical halon	- Use of water fire extinguishers on an electrical fire can cause life-threatening electrocution. - Dry chemical extinguishers leave a residue on surfaces that keeps them from igniting again. This residue is corrosive and often damages the items it protects from fire. - Carbon dioxide fire extinguishers do not leave a damaging residue.
	combustible metals (magnesium, titanium)	dry powder specific to the chemical that is burning	It is unsafe to use any non-specific fire extinguisher on a combustible metal fire.
	cooking media (vegetable and animal oils and fats)	wet chemical	- These are classified as cooking fires. They are of greatest concern in the food industry and restaurants. - Air-pressurized water extinguishers should not be used for these fires. They can spread the fire and cause burns from splattered fuel.



PULL THE PIN



AIM AT THE BASE



SQUEEZE TRIGGER



SWEEP

PASS is an acronym to help people remember the four steps to take when using a fire extinguisher.

Skills and Strategies

- Planning and Conducting
- Processing and Analyzing
- Evaluating
- Communicating

What You Need

- computer with Internet access

Is It Worth It?

Understandings and innovations of chemistry have resulted in the development of many synthetic products that have changed societies around the world. However, some of these products have had unintended consequences, such as a negative impact on the environment. In this investigation, you will choose a product to investigate and use information about its life cycle to answer the question, “Is it worth it?”

Question

What is the environmental impact of the production, use, and disposal of a synthetic product?

Procedure

1. In a group, choose one common synthetic product. Some options could include
 - a plastic product (such as a plastic water bottle)
 - an electronics product (such as a smartphone)
 - a synthetic fabric (such as clothing)
2. Develop a research plan based on finding out the following information about the chosen product.
 - its uses and the positive and negative contributions it makes to society
 - the technology and materials used to design and create it and the effects of the manufacturing processes (Is the extraction of raw materials required? Is there an environmental impact of that? Does processing of raw materials require large amounts of energy or other resources, such as water? Is it manufactured in a manner and place that supports workplace safety, equality, and healthy conditions?)
 - its expected life time (Has it been designed to be replaced often? to last for years?)
 - how it is discarded and whether it is biodegradable (Does it go to landfill? Does toxic waste result from its disposal?)
 - what recycling programs or “green” alternative products are available
 - answers to two questions about it that your group decides to investigate

Your plan should include deciding what sources of information you will use and how to share the work among group members so it is distributed equitably.

3. Have your teacher approve your plan before you proceed.
4. Carry out your research plan. As a group, combine your research findings to form a complete analysis.

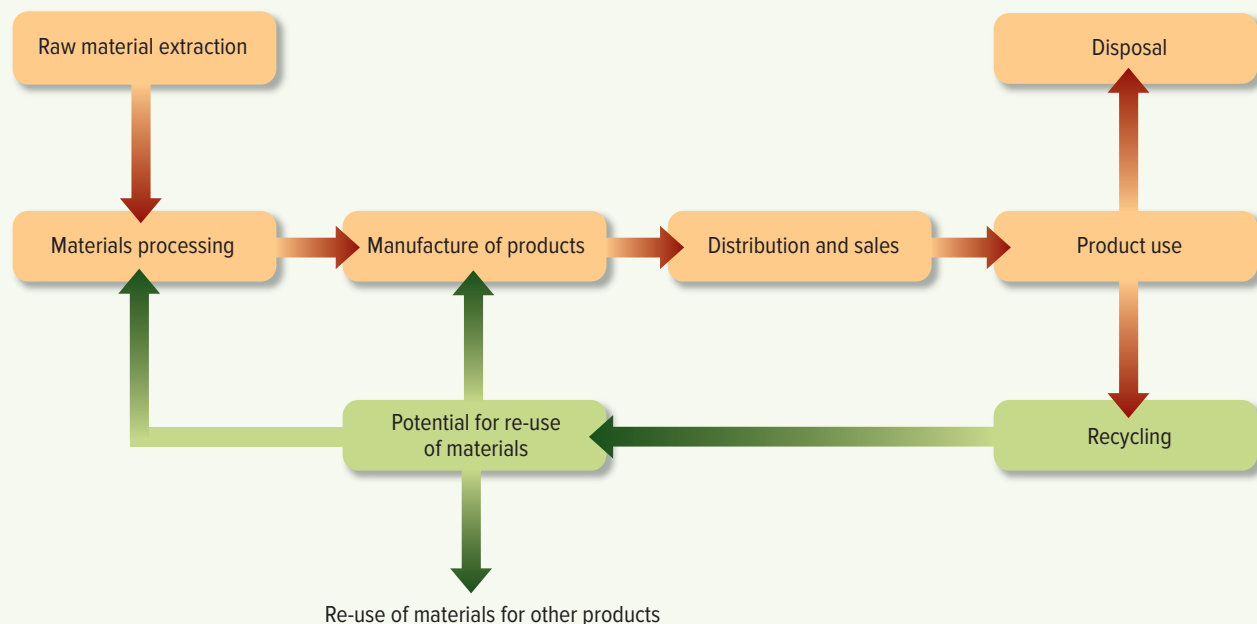
Process and Analyze

1. Describe the positive contributions that the product makes.
2. Describe the negative effects that the product has on the environment.
3. What are some ways that the negative consequences can be minimized by such things as “green” alternatives or recycling programs?

Conclude and Communicate

4. Share your findings and analysis with the class, using a format of your choice. Possibilities include a video documentary, social media website, or blog. Your presentation should include an answer to the question, “Is it worth it?”

A product’s life cycle includes all aspects involved in making, distributing, selling, using, and disposing of the product. The environmental and social impact at each stage of a product’s life cycle may be questioned and evaluated.



Skills and Strategies

- Planning and Conducting
- Processing and Analyzing
- Evaluating
- Communicating

What You Need

- computer with Internet access

Prepare an Emergency Response Plan

Chemicals are part of our everyday lives but accidents can threaten the well-being of people and the environment. Being prepared for possible emergencies helps to reduce the effects of accidents. In this investigation, you will develop an emergency response plan for a hypothetical situation.

Question

What features make an effective emergency response plan for a school or community?

Procedure

1. Working in small groups, decide on the type of emergency response plan you want to develop. For example, is it for a school? a small community? a large city? a workplace?
2. Choose one scenario for your emergency plan. Some examples are provided below.
 - An earthquake causes a gas leak in the school while school is in session.
 - An earthquake causes a gas leak in your community.
 - A transportation accident causes a dangerous chemical spill on a busy highway near a large city.
 - A fire breaks out in the school and the fire alarm goes off.
 - A fire breaks out in your home during the night.
3. Include the following considerations in your plan:
 - The preparation and planning required to be ready for an emergency. This includes the people who should be involved in organizing and executing the plan.
 - What the evacuation procedure will be and under what conditions it is to be followed
 - How the emergency response plan will be communicated to people; what the lines of communication will be among people involved in organizing and executing the plan
 - How any contamination is to be dealt with and cleaned up; how affected material is to be disposed of

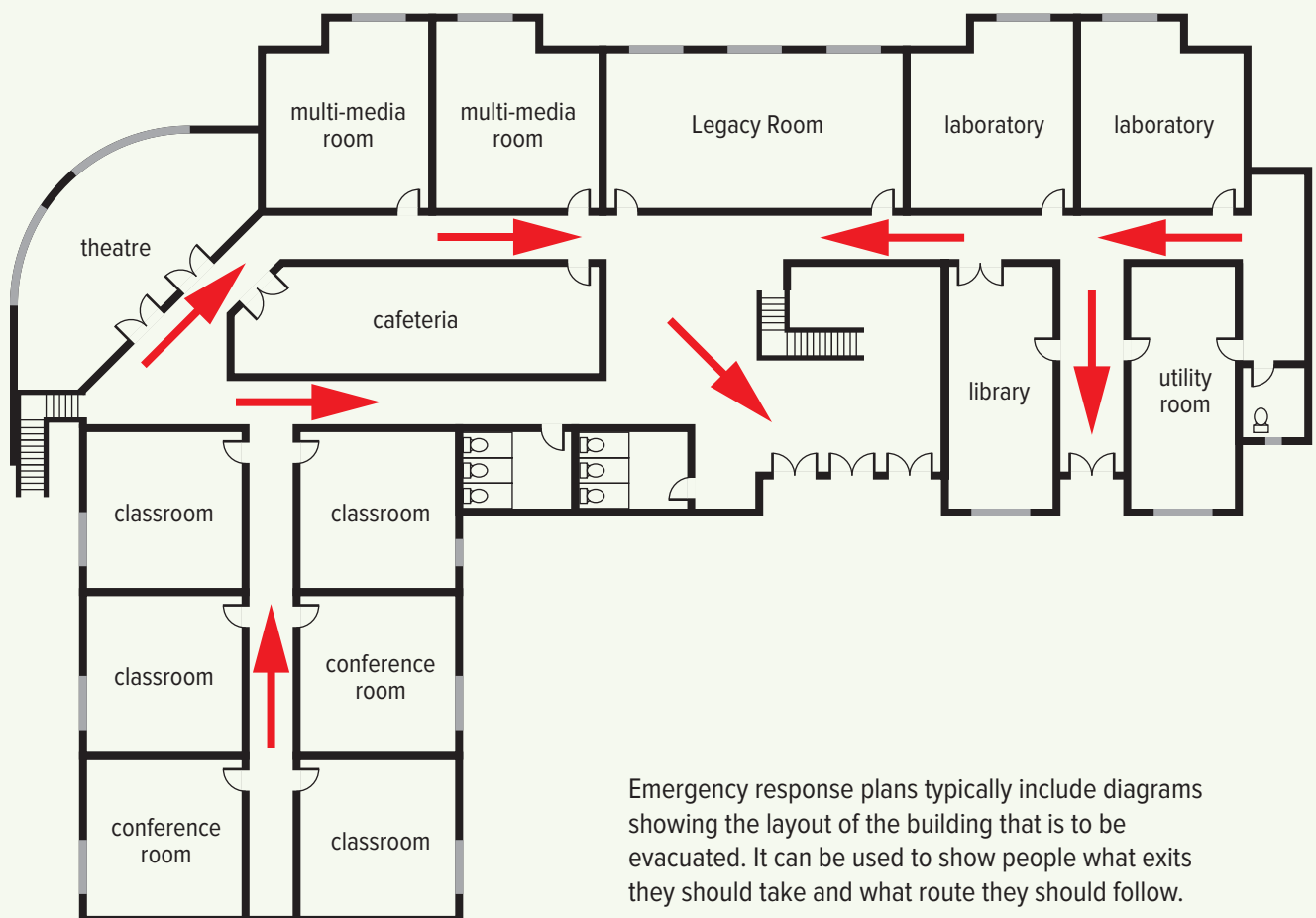
4. With your teacher's permission, contact local first responders. Find out some things they consider when developing an emergency response plan.
5. Develop your emergency response plan.

Process and Analyze

1. Find out what emergency response plan and protocols your school's Health and Safety Committee has developed. In terms of the different factors that were considered, how does it compare with your group's plan?

Conclude and Communicate

2. Share your emergency response plan with the class. Then develop a presentation, using the format of your choice, that could be used to convey this information to the school, town, or city involved. Keep your audience in mind when designing your presentation.



Emergency response plans typically include diagrams showing the layout of the building that is to be evacuated. It can be used to show people what exits they should take and what route they should follow.